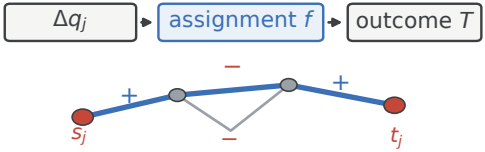
the same pipeline serves reference and learned assignment maps

learned assignment → edge attribution → topology certificate → adaptive summary → response supervision

(a) The explanation object

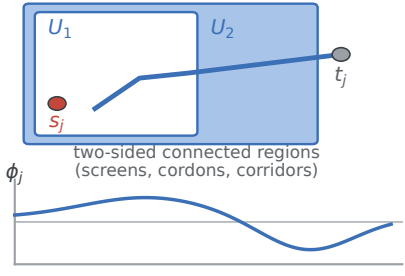


signed entries; other ODs' rerouting enters too

$$c_j = \Delta q_j \int_0^1 \mathrm{diag}(\nabla T(f)) J_f \mathbf{e}_j \, d\lambda$$

$C_j = \mathbf{1}^\top c_j$ is the system-wide contribution

(b) Nested boundaries and the profile



$\phi_j(U) = -\mathbf{1}_U^\top B c_j$, signed sum on crossing links

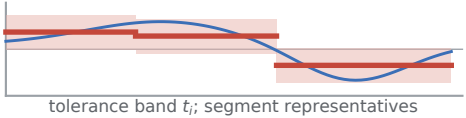
(c) Adaptive summary

one value suffices

$S_j^F \le 2\varepsilon$:
report $\beta_j^F \pm \varepsilon_j^F$

profile needed

minimum segments,
provably fewest



Anaheim: 83−211 boundary values → 4−14 segments at the working tolerance

(d) Learned assignment

structural certificate
weights constant on the traversed block atoms
(Theorem, per state)

exact validation
node divergence gives
$\max_U |\phi_j^L - \phi_j^S|$ exactly

response supervision
trains the flow Jacobian identified by the error bound

summary error $\le \varepsilon_j^{\mathrm{sum}} + \delta_j$
(compression + model error)

compatibility and compression are certified from the learned model alone; learned-reference fidelity is an offline validation guarantee